Combined exact gate software controls

Task T-1427. 3 November 2026.

Exact reviewed gate source	Source and recipe independently reviewed
Parser and host checks	45 checks pass; pristine candidate exits126
Native helper controls	154 checks pass; uid/gid substituted
Runtime / image admission	False; no candidate promoted

Candidate rejects nonPID1; combined software controls pass; runtime qualification pending. T-1427.

The single frozen attempt passed all 45 parser and host-negative checks, all 154 held-release helper checks, and the pristine candidate non-PID1 rejection control. The exact candidate returned 126 with the expected PID/UID/GID boundary diagnostic. Total elapsed time was 15.807886 seconds; the invalid-token wait control took 15.001871 seconds. Every launched child was reaped without uncertainty or pending signals.

The gate source remains SHA256 6548d8d41a37975bb30c1a225583113ea6c25056753b3d455d90f1608d8e65bc. Candidate SHA256 830406f4c2ddaf900ff4f95ca85f5b1c111b71c9d4d8bf6467ffa46853ae88e9 is the previously built static x86-64 binary; it was copied unchanged and ELF-inspected. Unlike the prior build packet, this packet intentionally executes the pristine candidate once as an unprivileged non-PID1 process to verify rejection before privileged observations or tested program execution.

The prior 45-check source was copied with only its include changed to the exact held gate. Its mountinfo open operation is redirected to pinned fixture bytes. Nonce syntax, mount mapping, bounded readers, exact values, malformed arguments and non-PID1 main rejection are covered. The 154-check source is unchanged: only fstat uid/gid are interposed for unprivileged fixtures; file type, mode, size, bytes and link count remain real. It covers 37 partial-file lengths, complete tokens, invalid metadata and tokens, and bounded invalid-token waiting.

The recipe verified all 364 compiler file hashes and resolved paths, and all frozen sources and fixtures, before execution. Limits were one attempt, 1 GiB address space, 175 seconds to TERM plus five seconds kill reserve, at most 60 seconds per compile and 25 seconds for the release helper. Explicit child environment and process-group termination/reaping were retained. No Docker, systemd, namespaces, mounts, root or cloud operations occurred.

The successful control receipt uses the qualifier's required software status only after these actual checks passed. It binds the exact gate source and binary, held-release success, proposal/admission digest, and terminal process/signal conditions. A non-author reviewer inspected the frozen recipe before execution and the completed evidence afterward; the reviewer ran no binaries. This completes the fixed software-control dependency. Actual kernel/container qualification, root ownership and Docker transport remain unvalidated; runtime and image admission remain false.